Project Documentation Format

1. Introduction

- **Project Title:** Import Data using Transform Maps (Spreadsheet)
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2. Project Overview

- **Purpose:** To import data from an Excel spreadsheet into a ServiceNow table quickly and accurately using an Import Set table and a Transform Map, without creating duplicate records.
- **Features:** Spreadsheet and custom table creation; Import Set table; Transform Map with field maps; data transformation and validation; coalesce to avoid duplicates; re-import of new data in Excel format; reports and dashboard.

3. Architecture

- **Data Source Layer:** Excel spreadsheet attached to a Data Source and loaded into ServiceNow.
- **Staging Layer:** Import Set table that holds the raw spreadsheet rows.
- **Transformation Layer:** Transform Map with field maps, validation and coalesce.
- **Database:** Custom target table in the ServiceNow database that stores the clean records.
- **Presentation Layer:** Reports and a dashboard built on the target table.

4. Setup Instructions

- **Prerequisites:** ServiceNow instance (for example a Personal Developer Instance), a user with admin / import roles, and a prepared Excel (.xlsx) file.
- **Installation:** No installation is needed. Log in to the instance, create the custom table, and prepare the spreadsheet with column headers that describe each field.

5. Application Structure

- **Spreadsheet:** Excel file with one header row and data rows.
- **Target Table:** Custom table with the required fields (Milestone 1).
- **Import Set Table and Transform Map:** Staging table and mapping from spreadsheet columns to target fields (Milestone 2).
- **Reports and Dashboard:** Report 1 and the dashboard that displays it (Milestone 4).

6. Running the Application

- Open System Import Sets > Load Data, choose the target import set table, upload the Excel file and load the data.
- Open the Transform Map, confirm the field maps and the coalesce field, and choose Transform.
- Open the target table to verify the records and check the Transform History for errors.
- To import new data, add new rows to the Excel file and repeat the load and transform steps.

7. API Documentation

- No custom API is built in this project; the import is run through the ServiceNow user interface.
- Optionally, the ServiceNow Import Set API can be used to send rows to the staging table and apply the same transform map.

8. Authentication

- Users log in to the ServiceNow instance with their credentials.
- Authorization is role based: admin or import roles are needed to load data and run transform maps; ACLs on the target table control who can read and write records.

9. User Interface

- Standard ServiceNow interface: Load Data page, Import Set rows list, Transform Map form, target table list view, Report designer and Dashboard page.

10. Testing

- Load the spreadsheet and confirm that the number of import set rows equals the number of spreadsheet rows.
- Run the transform and confirm that records appear in the target table with correct field values.
- Re-import the same file with coalesce enabled and confirm that no duplicate records are created.
- Add new rows to the Excel file, import again, and confirm that only new records are inserted.
- Open the report and dashboard and confirm that the counts match the target table.

11. Demo

- The demo link and GitHub link for the project are added on the Project Progress page of the project workspace.

12. Known Issues

- If spreadsheet column names change, the field maps must be updated.
- If the coalesce field contains blank or non-unique values, records may be updated incorrectly.
- Invalid data types in the spreadsheet (for example text in a date column) cause row errors during transform.

13. Future Enhancements

- Add transform scripts for advanced validation and data clean-up.
- Schedule imports automatically using a scheduled data import.
- Add more reports and dashboards and notifications for failed rows.